ISBA 4775-01: Networking & Cloud Computing

Loyola Marymount University | Fall 2026

Course	ISBA 4775-01, Networking & Cloud Computing
Credit hours	4
Meets	Tuesday and Thursday, 1:45 PM to 3:25 PM
Location	Hilton 113

Instructor

Instructor	Greg Lontok, Clinical Associate Professor
Office	Hilton 114
Preferred contact	Microsoft Teams message, for the fastest response
Course Teams channel	https://r.isba.co/networking-cloud-teams-fa26
Email	gregory.lontok@lmu.edu
Office hours	By appointment. Book at https://calendly.com/greg-lontok

Course description

This course will introduce students to the fundamentals of networking and cloud computing. Students will configure networks and cloud computing services to address common information systems and business analytics needs through hands-on exercises. The course will start with networking fundamentals covering topics such as networking hardware, the OSI Model, TCP/IP, and various network protocols, while addressing each topic's security considerations. The second half of the course will leverage the student's networking foundation to explore and deploy the most commonly used cloud services, including compute, storage, databases, and serverless. Students will be able to host cloud-based applications and know the difference between the various cloud services available. This course may not be repeated for degree credit.

Prerequisites

ISBA 3710 with a minimum grade of C-.

What this course is for

You already know how to code, and this course is about what happens after that: getting something you built to run somewhere other than your laptop, keeping it running, and being able to explain how it works to someone who asks hard questions.

By the end you should be able to walk into an unfamiliar system and make sense of it.

What you own

You will use AI throughout this course, heavily, and that is expected.

What stays yours is the system. You know how it works, how it talks to other systems, what it is for, and how it delivers value to the stakeholder. It has to be fit for purpose: not only that you can build one, but that it suits the environment and constraints it has to live in.

Two whiteboard interviews, together worth 42.5% of your grade, are where that gets observed. You diagram your own system from memory, with no notes and no repository open, and then answer for it. "Working with AI" below is the full policy.

Learning outcomes

By the end of this course you can:

1. Read an unfamiliar system. Identify its components, dependencies, and boundaries, and ask the questions that reveal purpose, alternatives, cost, and where the pieces are isolated or coupled.
2. Design a system fit for purpose. Select and justify infrastructure against a stakeholder's real environment, constraints, and budget, and defend why that choice differs for a hobby project, a startup, and an enterprise.
3. Build and deploy an application on cloud infrastructure, moving it deliberately through local development, staging, and production.
4. Troubleshoot a system under failure. Form and test hypotheses, isolate the fault along the dependency chain, and describe what went wrong in the language practitioners use.
5. Operate a deployed system responsibly for access, security, performance, and cost, and produce evidence it is working as intended.
6. Explain and defend the system to a stakeholder. What it does, why it was built this way, which alternatives were rejected, where its limits are, and what should happen next.

Grading

Component	Points
Exercises (10 × 15)	150
Project 1, Own Your Corner of the Internet	125
Midterm whiteboard interview	125
Project 2, the job-description-driven build	300
Final whiteboard interview	300
Total	**1000**

Projects are 42.5% and interviews are 42.5%. This course grades whether you can explain what you built about as heavily as the artifact itself.

Letter grades

Grade	Range		Grade	Range
A	93-100		C+	77-79
A-	90-92		C	73-76
B+	87-89		C-	70-72

Grade	Range		Grade	Range
B	83-86		D	60-69
B-	80-82		F	below 60

Exercises, 150 points

There are ten exercises, worth 15 points each. Exercise 01 and Exercise 02 are submitted as files through Brightspace. From Exercise 03 on, exercises go in your public `career-platform` repository, as specified in their requirements. Each one defines its required evidence and grading. Exercises are credit/no credit: credit earns 15 points and no credit earns 0. Meaningful attempts and honest investigation can earn credit even when a result needs correction.

Exercise 02 extends the in-class server lab: independently investigate a stopped Nginx service and a stopped MySQL service, explain your evidence, repair each service, and repeat the failed test. Submit a hand-drawn system diagram as a JPG and a short investigation report as a PDF. The two investigations, diagram and explanation, healthy baseline, and verified Codespace shutdown guide feedback within the credit/no-credit assessment.

Exercise schedule

#	Exercise	Type	Due
01	Accounts, your home network, and the self-discovery interview	Config	Tue 9/8
02	Troubleshoot a cloud service: Nginx and MySQL	Troubleshoot	Tue 9/15
03	Build it, then move it: spec, plan, and migration plan	Build	Thu 10/1
04	Domain delegated to Cloudflare DNS, live with HTTPS	Config	Thu 10/8
05	**Broken DNS/TLS: diagnose and fix**	Troubleshoot	Tue 10/13
06	Domain email and GA4: mail delivery, a page view, and an outbound click	Config	Tue 10/20
07	Migrate your site from Railway to AWS: EC2, RDS, verified cutover, and a diagram	Build	Tue 11/10, 1:45 PM
08	**Broken integration: an expired credential and a changed schema in Job Scout**	Troubleshoot	Tue 11/17, 1:45 PM
09	Raw agent loop in Python, three tools, evals, trace read	Build	Thu 11/19, 1:45 PM
10	**Broken agent: bad tool schema, runaway loop, cost blowup**	Troubleshoot	Thu 12/3, 1:45 PM

Ex01 and Ex02 were due at 1:45 PM, before class, and Ex03 is due at the same time on Thu 10/1. The Railway migration is a Project 1 requirement, taught on Thu 10/8. Exercise 05 is due Tue 10/13, after the DNS/TLS lesson. Exercise 06 is due Tue 10/20, covering the Zoho and GA4 lessons. Both are due at 1:45 PM. Basic GA4 is required for Project 1 and taught on Thu 10/15. Exercise 07 is the AWS migration, taught Thu 10/29 and Tue 11/3 and due Tue 11/10. Exercises 08 through 10 follow the Job Scout lessons, which start Tue 11/10.

Project 1: Own Your Corner of the Internet, 125 points

Project 1 runs from week 1 through week 8, and it is **due Thu 10/22 at 1:45 PM, before class.**

The exercises build the pieces. Project 1 is the integrated, documented, portfolio-grade assembly of them.

Deliverable	Requirement
Domain	Purchased, delegated to Cloudflare DNS, zone under your control
Site	Personal site on an Azure VM, then migrated to Railway with Railway-hosted PostgreSQL, live at your domain with valid TLS
Website update	Railway's GitHub autodeploy, with a reviewed and locally checked change, its commit and deployment, and verification of the live result
Domain email	Zoho Mail with MX, SPF, DKIM, and DMARC configured, plus verified sending, receiving, and replying from your domain
Analytics	GA4 with a verified page view and one useful interaction, plus an explanation of what each measures
Architecture diagram	Labeled services and data flow, including the migration and current system
Engineering record	A concise README and `docs/project-1-submission.md` linking configuration, plans, checks, and troubleshooting evidence already in the repository

The question is not whether you completed the steps but whether you would show this repository to an employer.

You may use your agent to reconstruct the engineering record from your files and Git history. Verify its account and link the supporting evidence. Rerun missing checks or mark them unverified, and identify accounts written afterward. The engineering record does not require a separate report.

Job Scout, Resend, and GitHub Actions move after the midterm. Exercise 07 is the AWS migration, taught October 29 and November 3 and due November 10. Job Scout starts November 10 and will help you find the posting for Project 2, with a vetted feed available if needed. Choose your Project 2 posting using the vetted feed if needed, so the agent doesn't block your first project milestone.

Midterm whiteboard interview, 125 points

The midterm is twenty minutes, in person and one-on-one, in Hilton 114. Slots run Monday, October 26, through Wednesday, October 28. Tuesday's interviews use our October 27 class period, and we meet for regular class on Thursday, October 29. Book one appointment:

https://calendly.com/greg-lontok/4775-midterm-interview-fa26

The interview is fifteen minutes of drawing, explanation, and follow-up questions, then a five-minute debrief. It covers our first-half class material, Exercises 01 through 06, and Project 1, from memory, without notes or your repository.

Draw how data travels from a device through network hardware to an application and back, including your original Azure setup and current Railway setup. Be ready to explain your choices, costs, troubleshooting checks, and evidence that the system works. The study guide has the scoring table, the minute-by-minute shape, and practice questions.

The changed constraint takes no more than three minutes. You'll know the categories beforehand, but the interview uses a different specific change: more traffic, a smaller budget, or a provider you must leave. Revise the affected part of your drawing and explain the tradeoff, verification, and recovery.

Most students have never done a whiteboard interview. The October one is where you find out what that is like while it still costs 125 points instead of 300, and that is why it sits before the November 13 withdrawal deadline.

Project 2: the job-description-driven build, 300 points

Project 2 runs from week 10 through week 15. You pick one posting your Job Scout surfaced, or use the vetted feed if needed, read the system that job implies, and build it. The architecture is fixed, and the job description chooses the problem domain.

Deliverable	Requirement
Web app	On a subdomain of the domain you already own
Agent	Custom-built: tools, memory, structured outputs, guardrails, embedded as an endpoint
Grounding	Domain data in S3, reached with least-privilege IAM credentials
Orchestration	A layer connecting two systems on a trigger, doing real work in the job's domain
Environments	Local, staging, production, with promotion through CI/CD
CI/CD	GitHub Actions, secrets managed properly, deploy on merge
Operations	Auth, rate limiting, analytics, cost controls, a monitoring story
Evidence	An eval set for the agent and an inspectable trace
Decision record	Your most important decision, the evidence, how AI contributed, what you accepted or rejected or changed, how you validated it, your biggest remaining uncertainty, and what is next
Blog post	The writeup

Everyone builds the same skeleton onto a different problem. A logistics posting might yield a shipment-status copilot, a marketing posting a campaign analyzer, and a healthcare-operations posting an intake triage assistant.

The orchestration requirement is a capability, not a tool. n8n, Make, Zapier, Power Automate, and code all count.

Your system has to be both deterministic and agentic, and you have to defend where the line between them falls. Knowing which parts should be a fixed workflow and which should be agentic is a design judgment, and it is the obvious curveball: your API costs tripled, so what moves out of the agent?

Milestones

These are gates inside the Project 2 grade rather than separate line items.

Milestone	Bar	Due
M1	Posting selected from your Job Scout run or the vetted feed, PRD written, architecture diagram approved	Tue 11/17

Milestone	Bar	Due
M2	App scaffolded and deployed to production on your subdomain	Tue 11/24
M3	Agent responding with tools, grounded in domain data via S3 and IAM	Thu 12/3
M4	Orchestration integrated, staging environment live and promoting to production	Tue 12/8
Final	Full system, operations hardened, evals, decision record, blog post	Thu 12/10

M4 fails without a working staging environment. The application runs fine without one, which makes it the first thing skipped under deadline pressure.

Final whiteboard interview, 300 points

Thursday, December 17, 11:00 AM, in the scheduled exam slot, for thirty minutes.

Same format as the midterm: twenty-five minutes on your own system from memory, five minutes of curveball. The system you defend is the one you built for a job you chose, so this is a simulated interview for that job.

Semester at a glance

Weeks	What happens
1-8	Networking foundations, then the cloud arc. Exercises 01-06, Project 1
9	Midterm whiteboard interviews Monday through Wednesday, then AWS migration Thursday
10-15	Job Scout and Project 2, the job-description-driven build. Exercises 07-10, milestones M1-M4
Finals	Final whiteboard interview

Thanksgiving week meets once, on Zoom. Tuesday, November 24 is remote, and you join from the Zoom tool in Brightspace rather than coming to Hilton 113. Thursday, November 26 is Thanksgiving Day, and it is the only cancelled meeting all semester.

Labor Day and Autumn Day fall on days this course doesn't meet.

Key dates

Event	Date
Add/drop without a W	Fri Sep 4
Project 1 due	Thu 10/22
Midterm interviews	Mon 10/26 to Wed 10/28
Last day to withdraw	Fri Nov 13
Instruction ends	Fri Dec 11
Final interview	Thu Dec 17, 11:00 AM

Class schedule

Date	Session	Before class	Due
Tue 9/1	The map: where a request goes, and the devices it passes through		
Thu 9/3	Finding things: host names, IP and MAC addresses, the default gateway, DNS, and what your own machine will tell you		
Tue 9/8	Ports and web services: requests, responses, HTTP status codes, and investigating a service failure		Ex01
Thu 9/10	Running and Troubleshooting Services: physical server, VM, container; Nginx and MySQL; processes, ports, logs; database creation, tables, and sample data		
Tue 9/15	Build your resume site: your own repository and Codespace, a specification and plan, a database-backed application, failure/recovery, and verified changes saved in Git		Ex02
Thu 9/17	Building with an agent and Git: executing the implementation plan, the request path through Nginx, the app server, and SQLite, and clone, add, commit, push, and pull		
Tue 9/22	Migrating to a VM: feature branches and pull requests, what Git moves and what it doesn't, IaaS, PaaS, and SaaS, and Azure resource groups		
Thu 9/24	Creating the VM: SSH keys, an Azure VM, a cloud firewall rule for SSH, and a written migration plan		
Tue 9/29	Finishing the migration: rebuilding the environment, moving live SQLite data, and verifying the app on the VM		
Thu 10/1	Operating the VM: Uvicorn workers and systemd, Nginx as a reverse proxy, recovering after a restart, and delegating your domain to Cloudflare DNS	Register your domain with the domain guide, a free .me or a $6.79 .com. Create a free Cloudflare account.	Ex03
Tue 10/6	HTTPS for your domain: certificates, the request path, renewal, and troubleshooting DNS/TLS	Run `nslookup -type=NS yourdomain` and confirm Cloudflare's name servers. Confirm your site loads over HTTP. No Resend signup is needed.	

Date	Session	Before class	Due
Thu 10/8	Migrating to a PaaS: PostgreSQL on Railway, comparing source and target data, cutting the domain over, and who manages what	By 10/7, finish Railway signup. Before class, complete Ex04 and the required Impeccable design activity. Keep Azure running until migration is verified.	Ex04
Tue 10/13	Zoho Mail at your domain: MX, SPF, DKIM, and DMARC, then sending, receiving, and replying	By 10/12, finish the Zoho account preparation below. Bring access to your domain's Cloudflare zone.	Ex05
Thu 10/15	What an MCP server is, and connecting your coding agent to Zoho Mail. Then GA4: page views, one useful interaction, and event verification	By 10/14, create your Analytics account, GA4 property, and web stream using the preparation below. Bring the measurement ID and access to your site repository.	
Tue 10/20	Project requirements review, practice interviews in pairs, and what a whiteboard interview is	Finish the site, migration, domain email, and GA4 checks. Bring your README and submission index. Draw the system from memory once.	Ex06
Thu 10/22	Guest speakers Cian O'Brien and Timothy Cherpes, ISBA alumni at ServiceNow	Submit Project 1 before class at 1:45 PM. By Wed 10/21 at 11:59 PM, read both LinkedIn profiles and post two questions in the Brightspace discussion.	Project 1
Tue 10/27	Midterm interviews replace class. More slots run Mon 10/26 and Wed 10/28.	Book one Calendly appointment and practice with the study guide.	
Thu 10/29	Migrate your site from Railway to AWS: code, data, configuration, and the domain	Bring access to your repository, Railway database, and Cloudflare zone.	
Tue 11/3	Migrate your site from Railway to AWS, continued: the AWS CLI, the deployment policy, the plan, RDS, and EC2	Install the AWS CLI and confirm you can sign in as your IAM user with MFA. Know where your Railway backup is.	
Thu 11/5	Professor Ken Summers: Troubleshooting Road Show, the full session	Keep working on Exercise 07.	
Tue 11/10	Job Scout 1: LLM calls, structured output, retrieval tools, and the first loop	By 11/9, complete the Claude API setup. Bring your `career-platform` repository.	Ex07
Thu 11/12	Job Scout 2: tools, state, stop conditions, a human approval step, a Resend digest, and integration failures	By 11/11, create a Resend account and bring Cloudflare access. Keep Zoho mail working.	
Tue 11/17	Reading your posting as a system spec, architecture review, then evals and trace reading on your Job Scout run	Finish Job Scout's first run. Choose your Project 2 posting from your run or the vetted feed, and bring your spec and architecture.	Ex08, M1
Thu 11/19	Environments and tested deployment: GitHub Actions deploying your AWS site, staging on a subdomain, and promotion through CI/CD	Bring eval cases and a reviewed application change. Check the tests locally.	Ex09

Date	Session	Before class	Due
Tue 11/24	Cloud storage and databases, least-privilege IAM, grounding and retrieval, and agent guardrails for loop limits, schemas, and costs. On Zoom	Bring your data-flow diagram and identify stored data, permissions, and the agent's stop conditions. Join through Brightspace's Zoom link.	M2
Thu 11/26	No class, Thanksgiving		
Tue 12/1	LMU Data Center tour with Jeff Solomon	Meet in University Hall in front of Einstein Bros. Bagels.	
Thu 12/3	Staging in practice, then the decision record	Complete the controlled agent-failure investigation and M3 checks.	Ex10, M3
Tue 12/8	Blog post workshop and final interview prep		M4
Thu 12/10	Final interview practice in pairs, and a course retrospective	Submit Project 2 before class at 1:45 PM.	Project 2

Professor Ken Summers leads the full class session on November 5, in Hilton 113, with his Troubleshooting Road Show. That evening, Brandon Kulik from Deloitte will speak on "Consulting in the Age of AI." The event runs from 6:00 to 7:00 PM in Hilton 107. RSVP at https://lmu.campuslabs.com/engage/event/12919719.

The October 6 revision keeps the Railway/PostgreSQL migration and focuses Project 1 on your website, domain, Zoho email, and basic GA4. October 20 is reserved for the project requirements review and interview practice. Job Scout, Resend, and GitHub Actions move after the midterm. Job Scout starts November 10, after the AWS migration, and the first Project 2 milestone follows it on November 17. Evals and tested deployment come next. No Resend signup or Claude API credit purchase is needed for Project 1.

Prepare before class

Complete account setup ahead of the lesson so we can use class time to configure and verify the system.

- **Before Thursday, October 8, at 1:45 PM:** complete the required website design activity with Impeccable alongside Ex04. Spend 30-45 minutes making two or three focused improvements in your existing repository, with before/after screenshots and checks. Follow the guide's fallback if installation or deployment is blocked. Bring the branch and honest results. This preparation adds no separate grade item or points, and Ex04's HTTPS requirements remain in effect.
- **By Wednesday, October 7, for Thursday's migration:** sign in to Railway with GitHub and check your trial. The sign-in window is October 3-7, and the trial starts when you sign in. Create nothing yet. Finish Ex04, save your HTTPS and certificate-renewal evidence, and have the current Azure database ready to migrate. Keep the VM running until the app, data, and domain work over HTTPS on Railway.
- **By Monday, October 12, for Tuesday's email lesson:** create your own Zoho account using the Forever Free option on the Zoho Mail pricing page. Have your domain name and Cloudflare login ready. If Forever Free isn't offered in your signup region, contact me before choosing a paid plan or trial. Leave the mail DNS records for class, when we'll configure the mailbox and test delivery and replies together.
- **By Wednesday, October 14, for Thursday's analytics lesson:** sign in with a Google account and follow Google's Analytics setup instructions to create an Analytics account, a GA4 property, and a web data stream for your site's HTTPS URL. Bring the stream's measurement ID. Stop before installing the tag, which we'll do and verify together in class.
- **By Monday, November 9, for Tuesday's first Job Scout lesson:** complete the Claude API setup.

- **By Wednesday, November 11, for Thursday's second Job Scout lesson:** create a Resend account. We will configure Resend's sending subdomain in class, keeping Zoho's incoming-mail records intact. If you already created an account, keep it. There's no additional setup to do now.

Work load expectations

LMU follows the Carnegie Unit standard: one semester credit hour represents at least three hours of student engagement per week. See https://academics.lmu.edu/aprc/lmucredithourpolicy/

For this four-credit course that is an average of **12 hours per week** across the semester, including class meetings, exercises, project work, and preparation.

Expect this to be uneven. Weeks with a troubleshooting exercise and weeks around a Project 2 milestone run heavier than the middle of a build.

Required materials

There is no textbook. This course runs on real infrastructure, which costs real money, and most of the spending starts in Exercise 01.

What	Cost	When
Claude Pro or an accepted ChatGPT subscription	Claude Pro is $20/month on monthly billing; alternative plan pricing varies	Choose one in Ex01; follow the relevant assignment for coding-agent setup
Railway	The 30-day trial includes a one-time $5 credit and needs no card. After it ends, the Free plan's $1 a month may not keep your site running	Sign in between 10/3 and 10/7 for the Thu 10/8 migration
Domain name	Free for a year through the GitHub Student Developer Pack, otherwise $10 to $15	Before Thu 10/1, and it stays yours after the course
Cloudflare DNS	Free	Before Thu 10/1
GitHub	Free tier	Exercise 01
GitHub Codespaces	Personal-account compute/storage allowance; additional usage can cost money	Course Codespace in Session 04 for Ex02; new portfolio Codespace in Session 05
AWS	Free tier	Migration on Thu 10/29, then Project 2
Zoho Mail	Forever Free where offered; contact me if unavailable	Create your account by Mon 10/12 for Tue 10/13

What	Cost	When
Google Analytics 4	Standard account is free	Create the account, property, and web stream by Wed 10/14 for Thu 10/15
Resend	Free tier	Create an account by Wed 11/11 for Thu 11/12
Claude API	Billed separately from your subscription	Complete the setup guide by Mon 11/9 for Tue 11/10
Posting source or retrieval service	Depends on the approved source	Job Scout, after the midterm; no Google Cloud or Firecrawl signup is assigned now

Plan your budget around the AI subscription you choose, domain registration, and the services you actually run. Prices, credits, and eligibility can change. Check Claude pricing, Railway plans, and Codespaces billing before enabling paid usage. A free tier does not mean all usage is free.

Apply for the GitHub Student Developer Pack in week 1 and check the current offers and verification status in your account. Do not plan on a particular benefit until you have confirmed eligibility.

For the Codespaces lab, check your allowance with the instructor, keep forwarded ports private, and stop the environment when finished. Contact me if access is blocked or a billing change is requested. Bring cost questions to me early.

Required lab fees

None.

Instructional methods

We meet in person twice a week, and each session combines instruction, live work, and student walkthroughs of their own exercises. Guided setup and build-alongs include predictions, live troubleshooting, and brief discussion. Detailed incident reports and diagrams are completed independently in the assigned homework. There are also two one-on-one whiteboard interviews.

Assignments and feedback

Follow each assignment's submission instructions. Ex01 and Ex02 go to Brightspace as separate files. Later builds use repositories and live systems as specified in their requirements; projects include their repositories and deployed systems. Due dates are announced in Brightspace and reflected in this syllabus. A date marked proposed or to be announced is not a confirmed deadline.

For Ex02, Google Docs saves your report automatically. Confirm it is saved to Drive, stop the Codespace, add your shutdown note, and download the report as a PDF for submission. Your diagram is a separate JPG. The Codespace should be stopped after testing; it does not need to remain running for grading.

Feedback is given in class, in Brightspace, or in the repository, depending on the assignment. Interview feedback is given verbally at the end of the interview.

Projects lose 10% per day when late, down to a floor of 50%. Percentage late penalties apply only to projects. Exercises remain credit/no credit, 15 or 0 points, including late work. Finish and submit the work even when it is late.

If you know you're going to miss a deadline, send me a Teams message before it passes. I may move it. That's case by case, and there's no bank of late days to spend. For projects, the 10% per day runs from the new date if I move it. Otherwise it runs from the original project deadline. Exercises retain their 15-or-0 grading without a percentage late deduction.

Some assignments are delivered in person at a fixed time, and the meeting time is the deadline. The two whiteboard interviews can't be rescheduled, so missing one scores zero on it.

Emergencies sit outside all of this. Tell me when you're able and we'll make a plan.

Attendance

Your scheduled commitments are class sessions, your midterm interview slot, and the final interview.

If you feel ill, stay home. Rest, and keep everyone else healthy.

When you can't make one, send me a Teams message. Beforehand is best, and as soon as you can if something happens that morning. Late still beats silent.

Your first three absences cost you nothing if you told me. You don't owe me a reason and I won't ask for documentation.

An absence you never tell me about costs 2% of your total grade, and it still uses up one of the three. From the fourth on, each absence costs 2% whether you told me or not. That 2% comes off your final grade after everything else is totaled, so attendance is not a line in the grading table.

Emergencies sit outside all of this. Serious illness, hospitalization, a death in the family: the three-absence count doesn't apply. Tell me when you're able and we'll work out a plan for the rest of the term.

Use of technology

We introduce tools when the relevant lesson or assignment needs them. In Session 04, students create a Codespace directly from the course repository. They install services and create a MySQL database with sample data.

In Session 05, students create a public `career-platform` repository and open their own Codespace using the resume-site guide. They start with a README and Python `.gitignore`, then use the coding agent to develop their specification and implementation plan before building.

The coding agent interviews students, then helps build a Python application after plan approval. Students verify database reads, failure, and recovery. They commit and push a working version, then make and verify a small change before committing and pushing again.

The early server lab does not require a local editor, database, or coding-agent installation. Follow each later lesson guide for its workflow setup. Cloudflare, Railway, Zoho Mail, and GA4 support Project 1. Resend, AWS, and orchestration tools follow after the midterm. The class schedule's Before class column and preparation checklist list what to set up ahead of each meeting.

For technology assistance, contact the ITS Service Desk:

- Phone: 310-338-7777 or 213-736-1097
- Email: servicedesk@lmu.edu
- Self-service: https://its.lmu.edu/servicedesk

Working with AI

Use AI for everything it's good at: planning, writing code, debugging, reading unfamiliar documentation, explaining errors, analyzing data, and drafting documents. There's no limit on how much you use it, and there is no requirement to log your prompts. Every assignment assumes you have a capable AI tool open beside you while you work.

What stays yours is the understanding. You should be able to explain anything an AI built for you to someone who asks hard questions, with no notes and no tool open. In this course that someone is me, twice, at the whiteboard. In another course it might be a presentation, a conversation in office hours, or an exam, and the standard is the same in each.

Four habits make that possible, and we practice them in class.

1. Ask before you build, which means having the tool read the actual documentation, data, or code before it proposes anything. A tool that guesses produces work you will have to undo.
2. Review the plan before you approve it. The tool writes a plan, you read it and change what is wrong, and only then does it build.
3. Verify what it claims, since a tool can report success on something that didn't happen. Check the result yourself, live, where it runs, instead of trusting the tool's own summary.
4. Record what happened: what you decided, what the tool recommended that you rejected or changed, what you tested, and what you haven't verified. Saying "not verified" is an honest answer that earns credit, while a claim you can't back up does not.

Two things never go into the tool: your secrets, meaning passwords, API keys, and account numbers, and anything that identifies another person. Keep both out of prompts, chat logs, and Git.

I grade understanding and honesty rather than polish. A rough result with a clear record of what you tried and what you learned earns credit. A finished result you can't explain does not. The only academic honesty issue with AI in this course is misrepresentation. That means claiming you verified what you didn't, or presenting an understanding you don't have as your own.

Independent work is not the same as independent thinking. You can use AI heavily and demonstrate excellent ownership, or work entirely alone and demonstrate very little.

Student support

- Academic Resource Center: https://academics.lmu.edu/arc/
- Student Psychological Services: https://studentaffairs.lmu.edu/wellness/studentpsychologicalservices/
- Disability Support Services: https://academics.lmu.edu/dss/
- Community of Care: https://studentaffairs.lmu.edu/lmucares/

If something outside this course is affecting your ability to do the work, tell me.

University policies

Academic honesty

Loyola Marymount University is a community dedicated to academic excellence, student-centered education, and the Jesuit and Marymount traditions. As such, the University expects all members of its community to act with honesty and integrity at all times, especially in their academic work. Academic honesty requires that all members of the LMU community act with integrity, respect their own intellectual and creative work as well as that of others, acknowledge sources consistently and completely, act honestly during exams and on assignments, and report results accurately. As an LMU Lion, by the Lion's Code, you are pledged to join the discourse of the academy with honesty of voice and integrity of scholarship.

Academic dishonesty will be treated as an extremely serious matter, with serious consequences that can range from receiving no credit for assignments/tests to expulsion. It is never permissible to turn in any work that has been copied from another student or copied from a source (including Internet) without properly acknowledging/citing the source. It is never permissible to work on an assignment, exam, quiz or any project with another person unless your instructor has indicated so in the written instructions/guidelines. It is your responsibility to make sure that your work meets the standard of academic honesty set forth in the "Academic Honesty Policy" found at: https://academics.lmu.edu/honesty/ For an additional resource, see and the "LMU Honor Code and Process" found at: https://bulletin.lmu.edu/content.php?catoid=1&navoid=18#LMU_Honor_Code_and_Process

What this means in this course

AI use is expected and encouraged here, on the terms in "Working with AI" above. Submitting a system you cannot account for is a violation regardless of what produced it.

Two more violations are specific to this course. The first is presenting a system as working without evidence that it works. Report actual observations, including failed checks and unresolved problems; do not fabricate output. Evidence may be a private report, a repository, or a live demonstration, as specified in the assignment. The second is presenting someone else's system, drawing, or evidence as your own. You must be able to explain what you submit.

Special accommodations

The DSS Office offers resources to enable students with physical, learning, ADD/ADHD, psychiatric disabilities and those on the autism spectrum to achieve maximum independence while pursuing their educational goals. Staff specialists interact with all areas of the University to eliminate physical and attitudinal barriers. Students must provide documentation for their disability from an appropriate licensed professional. Services are offered to students who have established disabilities under state and federal laws. We also advise students, faculty and staff regarding disability issues. Students who need reasonable modifications, special assistance, academic accommodations or housing accommodations should direct their request to the DSS Office as soon as possible. All discussions will remain confidential. The DSS Office is located on the 2nd floor of Daum Hall and may be reached by email at dsslmu@lmu.edu or phone at (310) 338-4216. Please visit http://www.lmu.edu/dss for additional information.

The whiteboard interviews can be accommodated. The earlier I know, the better I can plan them.

Tentative nature of the syllabus

This syllabus and its contents are subject to revision; students are responsible for any changes or modifications announced or distributed in class or posted on LMU's course management system.

Exercise scope and dates may shift as the semester develops. Any revision will be announced in class and through Brightspace, the same channels through which this syllabus was distributed. The version at https://r.isba.co/networking-cloud-syllabus-fa26 is always current, and its revision history is visible in git.

Classroom conduct and communication

Treat your classmates with professional respect, particularly when walking through each other's work. Much of this course involves showing something broken in front of the room, which only works if that is a safe thing to do.

Check your lion.lmu.edu email and Microsoft Teams regularly. Those are the channels I use and you are responsible for what is sent through them.

LMU's expectations for classroom behavior are at https://lmu.box.com/s/v2x89uspgbx3l23egcz7mjd6dbekcn60

Reporting requirements for sexual or interpersonal misconduct: https://studentaffairs.lmu.edu/lmucares/

Emergency preparedness information: https://www.lmu.edu/emergency

College of Business Administration mission

We advance knowledge and develop business leaders with moral courage and creative confidence to be a force for good in the global community.

https://cba.lmu.edu/about/mission/